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VCG Mechanism:

↳ Pivotal Mechanism *

→ N agents, designer uses VCG to select outcome s.t.

$$x^*(\theta) \in \operatorname{argmax}_{\theta} \sum_i v_i(x_i, \theta_i)$$

$$t(\theta) = \max_{x \in X} \sum_{j \neq i} v_j(x, \theta_j) - \sum_{j \neq i} v_j(x^*(\theta), \theta_j)$$

↳ Example:

	A	B	A+B
1	50	70	100
2	100	50	120
3	80	80	160

$$x^*(2) = A; \quad x^*(3) = B$$

$$t(1) = 0$$

$$t(2) = 0 + 80 - 160 = -80$$

$$t(3) = \underline{100} - (100 + 70) = -70$$

Read Thm 6.4:

↳ path-connected: 2 points in a set can be connected w/in set.

↳ Any strategy which uses the efficient decision rule in dominant strategies is a VCG

↳ when reading proofs, find where assump. are used.

More Properties of VCG:

↳ Prop 6.5: given a VCG (pivotal)^①, if values over all outcomes ≥ 0 , then $\sum$ transfers are ≥ 0

→ no negative externalities, designer never subsidizes

$$\sum t_i(\theta) = \sum_{i=1}^N \max_{x \in X} \sum_{j \neq i} v_j(x, \theta) - \sum_{i=1}^N \sum$$